

Physics 513 QFT HW 5

Pavel Okun

Department of Chemistry, University of Michigan, Ann Arbor USA

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CONTENTS

1. A	1
2. B	4
3. C	4

1. A

In my own preferred notation the retarded and advanced Green's functions, respectively, are

$$(2\pi)^4 \mathcal{G}_R(\vec{x}) = - \int d^4 p \frac{e^{-i\vec{p} \cdot \vec{x}}}{(p^0 + i\epsilon)^2 - p^2 - m^2} \quad (1)$$

$$(2\pi)^4 \mathcal{G}_A(\vec{x}) = - \int d^4 p \frac{e^{-i\vec{p} \cdot \vec{x}}}{(p^0 - i\epsilon)^2 - p^2 - m^2} \quad (2)$$

(3)

where I use arrows for 4-vectors and bold-face for 3-vectors and $p^2 = \mathbf{p}^2$.

$$E_p = \sqrt{p^2 + m^2}, \quad \vec{p} = (p^0, \mathbf{p}), \quad \vec{x} = (t, \mathbf{x}) \quad (4)$$

$$(2\pi)^4 \mathcal{G}_R(\vec{x}) = \int d^4 p \frac{e^{-i\vec{p} \cdot \vec{x}}}{E_p^2 - (p^0 + i\epsilon)^2} \quad (5)$$

$$(2\pi)^4 \mathcal{G}_R(\vec{x}) = \int d^3 p \int_{-\infty}^{\infty} dp^0 \frac{e^{-i\vec{p} \cdot \vec{x}}}{E_p^2 - (p^0 + i\epsilon)^2} \quad (6)$$

$$(2\pi)^4 \mathcal{G}_R(\vec{x}) = \int d^3 p \int_{-\infty}^{\infty} dp^0 \frac{e^{-i(p^0, \mathbf{p}) \cdot \vec{x}}}{E_p^2 - (p^0 + i\epsilon)^2} \quad (7)$$

$$(2\pi)^4 \mathcal{G}_R(\vec{x}) = \int d^3 p \int_{-\infty}^{\infty} dE \frac{e^{-i(E, \mathbf{p}) \cdot (t, \mathbf{x})}}{E_p^2 - (E + i\epsilon)^2} \quad (8)$$

$$(2\pi)^4 \mathcal{G}_R(\vec{x}) = \int d^3 p \int_{-\infty}^{\infty} dE \frac{e^{-i(Et - \mathbf{p} \cdot \mathbf{x})}}{E_p^2 - (E + i\epsilon)^2} \quad (9)$$

$$(2\pi)^4 \mathcal{G}_R(\vec{x}) = \int d^3 p e^{i\mathbf{p} \cdot \mathbf{x}} \int_{-\infty}^{\infty} dE \frac{e^{-iEt}}{E_p^2 - (E + i\epsilon)^2} \quad (10)$$

$$I_R(t, p) = \int_{-\infty}^{\infty} dE \frac{e^{-iEt}}{E_p^2 - (E + i\epsilon)^2} \quad (11)$$

$$(2\pi)^4 \mathcal{G}_R(\vec{x}) = \int d^3 p e^{i\mathbf{p} \cdot \mathbf{x}} I_R(t, p) \quad (12)$$

$$I_R(t, p) = \int_{-\infty}^{\infty} dx \frac{e^{-ixt}}{E_p^2 - (x + i\epsilon)^2} \quad (13)$$

I assume that $\epsilon \rightarrow 0_+$.

$$z_0 = -E_p - i\epsilon, \quad z_1 = E_p - i\epsilon \quad (14)$$

$$f_R(x) = \frac{e^{-ixt}}{(x - z_0)(x - z_1)} \quad (15)$$

$$I_R(t, p) = - \int_{-\infty}^{\infty} dx f_R(x) \quad (16)$$

I integrate along the contours in Fig. 1. I start by going

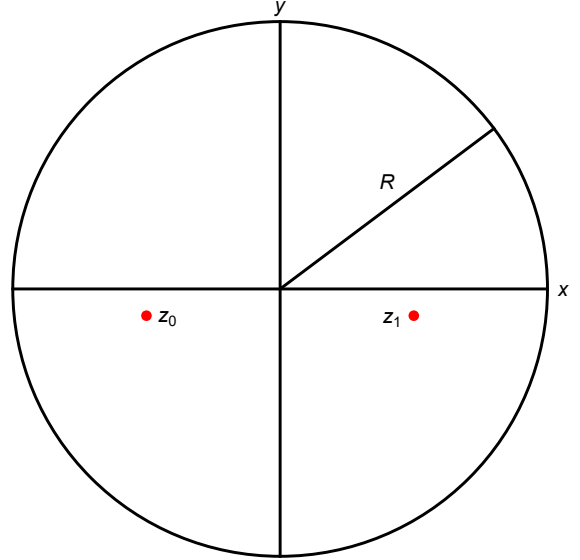


FIG. 1. Integration contours for I_R . I use the upper semicircle for $t < 0$ and the lower semicircle for $t > 0$. I assume $R \rightarrow \infty$.

along the lower semicircle for $t > 0$.

$$\oint dz f_R(z) = - \int_{-R}^R dx f_R(x) - \int_R^{-R} dz f_R(z), \quad R \rightarrow \infty. \quad (17)$$

The overall minus sign comes from integrating clockwise. I do the integral along the semicircle where $z = Re^{i\phi}$, and $dz = iRe^{i\phi}d\phi$.

$$\begin{aligned} \int_R^{-R} dz f_R(z) &= iR \int_0^{-\pi} d\phi e^{i\phi} f_R(Re^{i\phi}) = \\ &= iR \int_0^{-\pi} d\phi \frac{e^{i\phi} \exp(-iRe^{i\phi}t)}{(Re^{i\phi} - z_0)(Re^{i\phi} - z_1)} \end{aligned} \quad (18)$$

I take $R \rightarrow \infty$.

$$\begin{aligned} \int_R^{-R} dz f_R(z) &= \frac{1}{iR} \int_{-\pi}^0 d\phi \frac{\exp(-iRe^{i\phi}t)}{e^{i\phi}} = \\ &= \frac{1}{iR} \int_{-\pi}^0 d\phi \exp[-i(Rt \cos \phi + \phi)] = \\ &= \frac{1}{iR} \int_{-\pi}^0 d\phi \exp[Rt \sin \phi - i(Rt \cos \phi + \phi)] = \\ &= \frac{1}{iR} \int_{-\pi}^0 d\phi \exp(Rt \sin \phi) \exp[-i(Rt \cos \phi + \phi)] \end{aligned} \quad (19)$$

Because $t > 0$ and $\sin \phi < 0 \forall \phi \in (-\pi, 0)$ this integral is 0.

$$\begin{aligned} \int_R^{-R} dz f_R(z) &= \\ &= \frac{1}{iR} \int_{-\pi}^0 d\phi \exp(-R|t \sin \phi|) \exp[-i(Rt \cos \phi + \phi)] = \\ &= \frac{e^{-\infty}}{iR} \int_{-\pi}^0 d\phi \exp[-i(Rt \cos \phi + \phi)] = 0 \end{aligned} \quad (20)$$

$$\oint dz f_R(z) = - \int_{-R}^R dx f_R(x) - \int_R^{-R} dz f_R(z) \quad (21)$$

$$- \int_{-R}^R dx f_R(x) = \oint dz f_R(z) \quad (22)$$

$$I_R(t, p) = \oint dz f_R(z) \quad (23)$$

I evaluate the closed loop integral using the residue theorem and take the $\epsilon \rightarrow 0_+$ limit.

$$\oint dz f_R(z) = 2\pi i \frac{e^{-iz_0 t} - e^{-iz_1 t}}{z_0 - z_1} = \pi \frac{e^{iE_p t} - e^{-iE_p t}}{iE_p} \quad (24)$$

$$I_R(t, p) = \pi \frac{e^{iE_p t} - e^{-iE_p t}}{iE_p}, \quad t > 0 \quad (25)$$

For the $t < 0$ case I use the upper contour.

$$\oint dz f_R(z) = \int_{-R}^R dx f_R(x) + \int_R^{-R} dz f_R(z) \quad (26)$$

The upper contour encloses no residues, so

$$\oint dz f_R(z) = 0. \quad (27)$$

$$\oint dz f_R(z) = \int_{-R}^R dx f_R(x) + \int_R^{-R} dz f_R(z) \quad (28)$$

$$- \int_{-R}^R dx f_R(x) = \int_R^{-R} dz f_R(z) \quad (29)$$

$$- \int_{-\infty}^{\infty} dx f_R(x) = \lim_{R \rightarrow \infty} \int_R^{-R} dz f_R(z) \quad (30)$$

$$\begin{aligned} I_R(t, p) &= \lim_{R \rightarrow \infty} \int_R^{-R} dz f_R(z) = \\ &= i \lim_{R \rightarrow \infty} R \int_0^{\pi} d\phi e^{i\phi} f_R(Re^{i\phi}) = \\ &= i \lim_{R \rightarrow \infty} R \int_0^{\pi} d\phi \frac{e^{i\phi} \exp(-iRe^{i\phi}t)}{(Re^{i\phi} - z_0)(Re^{i\phi} - z_1)} = \\ &= i \lim_{R \rightarrow \infty} \frac{1}{R} \int_0^{\pi} d\phi \frac{\exp(-iRe^{i\phi}t)}{e^{i\phi}} = \\ &= i \lim_{R \rightarrow \infty} \frac{1}{R} \int_0^{\pi} d\phi \exp[-i(Rt \cos \phi + \phi)] = \\ &= i \lim_{R \rightarrow \infty} \frac{1}{R} \int_0^{\pi} d\phi \exp[Rt \sin \phi - i(Rt \cos \phi + \phi)] = \\ &= i \lim_{R \rightarrow \infty} \frac{1}{R} \int_0^{\pi} d\phi \exp(-R|t \sin \phi|) \exp[-i(Rt \cos \phi + \phi)] = \\ &= i \lim_{R \rightarrow \infty} \frac{e^{-\infty}}{R} \int_0^{\pi} d\phi \exp[-i(Rt \cos \phi + \phi)] = 0 \end{aligned} \quad (31)$$

$$I_R(t, p) = 0, \quad t < 0 \quad (32)$$

I combine Eqs. 25 & 32

$$I_R(t, p) = \pi \frac{e^{iE_p t} - e^{-iE_p t}}{iE_p} \Theta(t). \quad (33)$$

$$I_R(t, p) = \frac{2\pi}{E_p} \sin(E_p t) \Theta(t) \quad (34)$$

$$(2\pi)^4 \mathcal{G}_R(\vec{x}) = \int d^3 p e^{i\vec{p} \cdot \vec{x}} I_R(t, p) \quad (35)$$

$$(2\pi)^3 \mathcal{G}_R(\vec{x}) = \Theta(x^0) \int d^3p \frac{\sin(E_p x^0)}{E_p} e^{i\vec{p} \cdot \vec{x}} \quad (36)$$

$$(2\pi)^4 \mathcal{G}_A(\vec{x}) = \int d^4p \frac{e^{-i\vec{p} \cdot \vec{x}}}{E_p^2 - (p^0 - i\epsilon)^2} \quad (37)$$

$$(2\pi)^4 \mathcal{G}_A(\vec{x}) = \int d^3p e^{i\vec{p} \cdot \vec{x}} \int_{-\infty}^{\infty} dE \frac{e^{-iEt}}{E_p^2 - (E - i\epsilon)^2} \quad (38)$$

$$(2\pi)^4 \mathcal{G}_A(\vec{x}) = \int d^3p e^{i\vec{p} \cdot \vec{x}} I_A(t, p) \quad (39)$$

$$I_A(t, p) = \int_{-\infty}^{\infty} dx \frac{e^{-ixt}}{E_p^2 - (x - i\epsilon)^2} \quad (40)$$

$$z_2 = -E_p + i\epsilon, \quad z_3 = E_p + i\epsilon \quad (41)$$

$$f_A(x) = \frac{e^{-ixt}}{(x - z_2)(x - z_3)} \quad (42)$$

$$I_A = - \int_{-\infty}^{\infty} dx f_A(x) \quad (43)$$

I integrate using the contours in Fig. 2. I start with $t > 0$

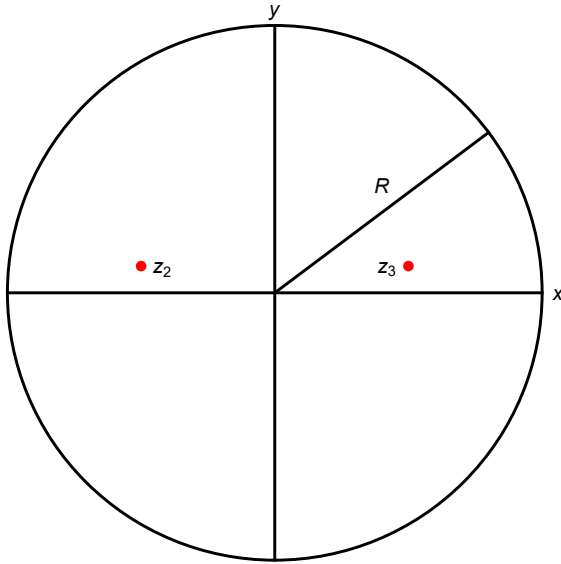


FIG. 2. Integration contours for I_A . I use the upper semicircle for $t < 0$ and the lower semicircle for $t > 0$. I assume $R \rightarrow \infty$.

along the lower semicircle.

$$- \oint dz f_A(z) = \int_{-R}^R dx f_A(x) + \int_R^{-R} dz f_A(z) \quad (44)$$

$$- \int_{-R}^R dx f_A(x) = \int_R^{-R} dz f_A(z) \quad (45)$$

$$- \int_{-\infty}^{\infty} dx f_A(x) = \lim_{R \rightarrow \infty} \int_R^{-R} dz f_A(z) \quad (46)$$

$$\begin{aligned} I_A &= \lim_{R \rightarrow \infty} \int_R^{-R} dz f_A(z) \propto \lim_{R \rightarrow \infty} R \int_{-\pi}^0 d\phi e^{i\phi} f_A(Re^{i\phi}) \\ &\propto \lim_{R \rightarrow \infty} R \int_{-\pi}^0 d\phi \frac{e^{i\phi} \exp(-iRe^{i\phi}t)}{(Re^{i\phi} - z_2)(Re^{i\phi} - z_3)} \propto \\ &\lim_{R \rightarrow \infty} \frac{1}{R} \int_{-\pi}^0 d\phi \exp(-iRe^{i\phi}t - i\phi) \propto \\ &\lim_{R \rightarrow \infty} \frac{1}{R} \int_{-\pi}^0 d\phi \exp(-iRe^{i\phi}t) \propto \\ &\lim_{R \rightarrow \infty} \frac{1}{R} \int_{-\pi}^0 d\phi \exp(-R|t \sin \phi|) \propto \\ &\lim_{R \rightarrow \infty} \frac{e^{-R}}{R} = \frac{e^{-\infty}}{\infty} = 0 \end{aligned} \quad (47)$$

$$I_A = 0, \quad t > 0 \quad (48)$$

For $t < 0$ I take the upper semicircle.

$$\oint dz f_A(z) = \int_{-R}^R dx f_A(x) + \int_R^{-R} dz f_A(z) \quad (49)$$

$$\oint dz f_A(z) = \int_{-\infty}^{\infty} dx f_A(x) + \lim_{R \rightarrow \infty} \int_R^{-R} dz f_A(z) \quad (50)$$

$$\oint dz f_A(z) = \lim_{R \rightarrow \infty} \int_R^{-R} dz f_A(z) - I_A \quad (51)$$

$$\begin{aligned} \lim_{R \rightarrow \infty} \int_R^{-R} dz f_A(z) &\propto \lim_{R \rightarrow \infty} R \int_0^{\pi} d\phi e^{i\phi} f_A(Re^{i\phi}) \propto \\ &\lim_{R \rightarrow \infty} \frac{1}{R} \int_0^{\pi} d\phi \exp(-R|\sin \phi|t) \propto \\ &\lim_{R \rightarrow \infty} \frac{e^{-R}}{R} = \frac{e^{-\infty}}{\infty} = 0. \end{aligned} \quad (52)$$

$$\begin{aligned} -I_A &= \oint dz f_A(z) = \pi \frac{e^{iE_p t} - e^{-iE_p t}}{iE_p} = \\ &\frac{2\pi}{E_p} \sin(E_p t) \end{aligned} \quad (53)$$

$$I_A(t, p) = -\frac{2\pi}{E_p} \sin(E_p t), \quad t < 0 \quad (54)$$

$$I_A(t, p) = -\frac{2\pi}{E_p} \sin(E_p t) \Theta(-t) \quad (55)$$

$$(2\pi)^4 \mathcal{G}_A(\vec{x}) = \int d^3 p e^{i\mathbf{p} \cdot \mathbf{x}} I_A(t, p) \quad (56)$$

$$(2\pi)^3 \mathcal{G}_A(\vec{x}) = -\Theta(-t) \int d^3 p \frac{\sin(E_p t)}{E_p} e^{i\mathbf{p} \cdot \mathbf{x}} \quad (57)$$

$$(2\pi)^3 \mathcal{G}_A(\vec{x}) = -\Theta(-x^0) \int d^3 p \frac{\sin(E_p x^0)}{E_p} e^{i\mathbf{p} \cdot \mathbf{x}} \quad (58)$$

2. B

$$\mathcal{G}_R(\vec{x}) - \mathcal{G}_A(\vec{x}) = [\Theta(x^0) + \Theta(-x^0)] \int d^3 p \frac{\sin(E_p x^0)}{(2\pi)^3 E_p} e^{i\mathbf{p} \cdot \mathbf{x}} \quad (59)$$

$$\Theta(x^0) + \Theta(-x^0) = 1 \quad (60)$$

$$\mathcal{G}_R(\vec{x}) - \mathcal{G}_A(\vec{x}) = \frac{1}{(2\pi)^3} \int d^3 p \frac{\sin(E_p x^0)}{E_p} e^{i\mathbf{p} \cdot \mathbf{x}} \quad (61)$$

In my Scalar Field notes I derived

$$[\phi(\vec{x}), \phi(\vec{y})] = \int d^3 p \frac{\sin[E_p(x^0 - y^0)]}{(2\pi)^3 i E_p} e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})}. \quad (62)$$

$$\mathcal{G}_R(\vec{x} - \vec{y}) - \mathcal{G}_A(\vec{x} - \vec{y}) = i \int d^3 p \frac{\sin[E_p(x^0 - y^0)]}{(2\pi)^3 i E_p} e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} \quad (63)$$

$$\mathcal{G}_R(\vec{x} - \vec{y}) - \mathcal{G}_A(\vec{x} - \vec{y}) = i[\phi(\vec{x}), \phi(\vec{y})] \quad (64)$$

3. C

$$S^{-1} \phi_{\text{in}} S = \phi_{\text{in}} + i \int d^4 y [\phi_{\text{in}}(\vec{x}), \phi_{\text{in}}(\vec{y})] j(\vec{y}) \quad (65)$$

Since only the in field appears in this problem I drop the subscript "in" from now on.

$$S^{-1} \phi S = \phi + i \int d^4 y [\phi(\vec{x}), \phi(\vec{y})] j(\vec{y}) \quad (66)$$

$$I = i \int d^4 y [\phi(\vec{x}), \phi(\vec{y})] j(\vec{y}) \quad (67)$$

$$S^{-1} \phi S = \phi + I \quad (68)$$

Equations 63 & 64 show that I is a c-number.

$$S = e^{-\sigma} \quad (69)$$

$$e^\sigma \phi e^{-\sigma} = \phi + i \int d^4 y [\phi(\vec{x}), \phi(\vec{y})] j(\vec{y}) \quad (70)$$

I use a variant of the Baker–Campbell–Hausdorff formula called the Campbell identity.

$$e^\sigma \phi e^{-\sigma} = \phi + [\sigma, \phi] + \frac{[\sigma, [\sigma, \phi]]}{2} + \mathcal{O} \quad (71)$$

$$\phi + [\sigma, \phi] + \frac{[\sigma, [\sigma, \phi]]}{2} + \mathcal{O} = \phi + I \quad (72)$$

$$[\sigma, \phi] + \frac{[\sigma, [\sigma, \phi]]}{2} + \mathcal{O} = I \quad (73)$$

This implies that $I = [\sigma, \phi]$ because I is a commutator.

$$I + \frac{[\sigma, I]}{2} + \mathcal{O} = I \quad (74)$$

Where the second and higher order terms vanish because I is a c-number and operators commute with c-numbers.

$$I = I \quad (75)$$

$$I = [\sigma, \phi] = i \int d^4 y [\phi(\vec{x}), \phi(\vec{y})] j(\vec{y}) \quad (76)$$

$$[\sigma, \phi(\vec{x})] = i \int d^4 y [\phi(\vec{x}), \phi(\vec{y})] j(\vec{y}) \quad (77)$$

$$-[\phi(\vec{x}), \sigma] = \left[\phi(\vec{x}), i \int d^4 y \phi(\vec{y}) j(\vec{y}) \right] \quad (78)$$

$$[\phi(\vec{x}), \sigma] = \left[\phi(\vec{x}), -i \int d^4 y \phi(\vec{y}) j(\vec{y}) \right] \quad (79)$$

$$\sigma = -i \int d^4 y \phi(\vec{y}) j(\vec{y}) \quad (80)$$

$$S = \exp \left[i \int d^4 y \phi(\vec{y}) j(\vec{y}) \right] \quad (81)$$

$$S = \exp \left[i \int d^4 x \phi_{\text{in}}(\vec{x}) j(\vec{x}) \right] \quad (82)$$